

Student's Guide

Algebra and Trigonometry

First -Term

First -year of Secondary

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Unit one

Algebra

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Unit one

Algebra

Dear student by the end studying of this unit you should have the following abilities and knowledge:

- ❖ To Solve the Quadratic Equations in One Variable Algebraically (Analytically) and Graphically
- ❖ *Recognize* the complex numbers (C) and the operation on it.
- ❖ To solve the second degree equation in the complex numbers.
- ❖ To find sum and difference between the two roots of second degree equation in the complex numbers
- ❖ To investigate the type of two roots of a quadratic equation
 - ❖ To investigate the sign of the function.
 - ❖ To solve quadratic inequalities in one variable

Evaluation Test for Newcomers

First question: Find the solution set of the following equations in R:

1. $x^2 = 4$

2. $x^2 - 3x - 10 = 0$

3. $-25 + 10x - x^2 = 0$

4. $3x - 15x^2 = 0$

Second question: Find the solution set of the following two equations in $R \times R$:

$x + y = 2$, $x - y = 6$

Third question: Find the numerical value of the expression:

$\sin 30^\circ \cos 60^\circ + \cos 30^\circ \sin 60^\circ$

Fourth question: Find $n(x) =$

$\frac{x^2 - 2x}{x^2 - 4} +$

$\frac{2x + 6}{x^2 + 5x + 6}$ in the simplest form,

show the domain of the function

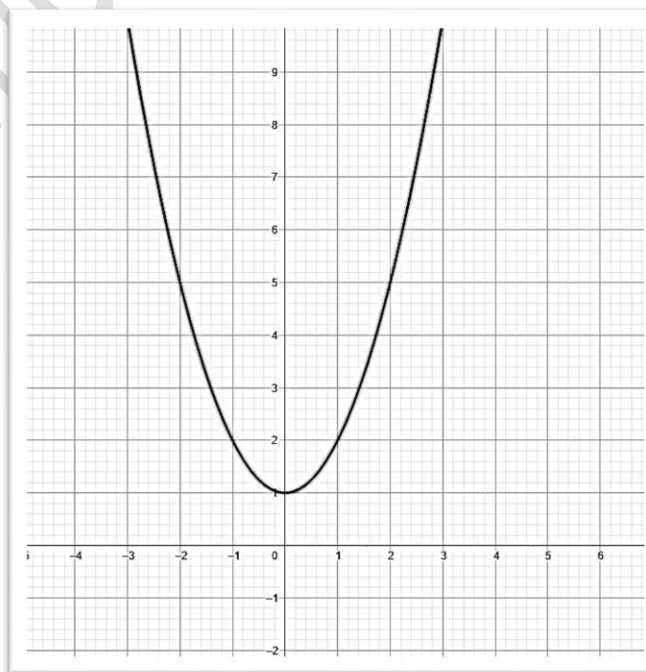
Fifth question: If the opposite figure represents the curve of a quadratic function f , then: the solution set of the equation: $f(x) = \text{zero}$ in R

1. $\emptyset$

2. $\{0\}$

3. $\{1\}$

4. $\{0, 1\}$



Revision

1. Factorization by tacking out the Highest Common Factor (H.C.F):

The expression = $2x^3 + 4x^2$

(Each of the two coefficients 2 and 4 is divisible by 2)

(The common factor of x^3 , x^2 is x^2 (with the smallest power))

The expression = $2x^3 + 4x^2 = 2x^2(x + 2)$

2. Factorization of the trinomial:

a) The expression = $x^2 + 5x + 6$ (the sign of the last term is positive)

(two numbers their product is 6 and their sum is 5)

The expression = $(x + 3)(x + 2)$ (the signs are the same as the middle term)

b) The expression = $x^2 + 5x - 6$ (the sign of the last term is negative)

(two numbers their product is 6 and their difference is 5)

The expression = $(x + 6)(x - 1)$

(the signs are different, the sign of the greatest one is the same as the middle term)

3. Difference between two squares:

The expression = $x^2 - y^2 = (x - y)(x + y)$

The expression = $x^2 - 4 = (x - 2)(x + 2)$

The expression = $x^2 - 9 = (x - 3)(x + 3)$

Notice that:

The difference between $\sqrt{4} = 2$ and $x^2 = 4, x = \pm\sqrt{4}$ $x = \pm 2$

4. Solving two equations from the first degree in two variables:

Find the solution set of the two equations: $x + y = 2, y = x + 2$ in $\mathbb{R} \times \mathbb{R}$

a) By substitution

$$x + (x + 2) = 2$$

$$2x = 0$$

$$x = 0$$

$$\therefore y = 0 + 2 = 2$$

$$S.S = \{(0,2)\}$$

b) By omitting

$$x + y = 2$$

$$-x + y = 2$$

$$2y = 4 \quad \therefore y = 2$$

$$x + 2 = 2 \quad x = 0$$

$$S.S = \{(0,2)\}$$

Lesson 1Solving the Quadratic Equations in One Variable
Algebraically (Analytically) and Graphically

❖ The general form of the second degree equation in one unknown is:

$$ax^2 + bx + c = 0$$

where:

- a, b are called coefficients of x^2 , x respectively, $a \neq 0$
- c is a constant number
- $\{a, b, c\} \subset R$.

First: Solving the second degree equation algebraically

1. Using factorization

Example 1:

Find the solution set of the equation : $x^2 - 1 = 0$ in R.

Solution:

$$x^2 - 1 = 0$$

$$(x-1)(x+1) = 0$$

$$x = 1 \quad \& \quad x = -1$$

$$S.S \{-1, 1\}$$

Example 2:

Find the solution set of the equation : $x^2 - 7x + 10 = 0$ in R.

Solution:

$$x^2 - 7x + 10 = 0$$

$$(x - 2)(x - 5) = 0$$

$$(x - 2) = 0 \text{ or } (x - 5) = 0$$

$$x = 2 \text{ or } x = 5$$

$$S.S. = \{2, 5\}$$

Example 3:

Find the solution set of the equation : $x^2 - 3x = 0$ in R.

Solution:

$$x(x - 3) = 0$$

$$x = 0 \text{ or } x = 3$$

$$S.S. = \{0, 3\}$$

Example 4:

Find the solution set of the equation : $x^2 - 8x + 16 = 0$ in R.

Solution:

$$x^2 - 8x + 16 = 0$$

$$(x - 4)^2 = 0$$

$$(x - 4) = 0$$

$$x = 4$$

$$S.S. = \{4\}$$

Example 5:

If $x = 3$ is one of the solutions of the equation $x^2 - 6x + k = 0$, then: $k = \dots\dots\dots$

$$x^2 - 6x + k = 0$$

substitute with $x = 3$ in the equation:

$$9 - 18 + k = 0$$

$$-9 + k = 0$$

$$k = 9$$

2. By using the general formula:

Example 6: Find the solution set of the equation

$$x^2 - 3x + 1 = 0 \text{ in } \mathbb{R}$$

Solution:

$$a = 1, b = -3, c = 1$$

$$x = \frac{-b \pm \sqrt{b^2 - 4ac}}{2a}$$

$$x = \frac{3 \pm \sqrt{9 - 4}}{2}$$

$$x = \frac{3 + \sqrt{5}}{2} \cong 2.62 \quad \text{or} \quad x = \frac{3 - \sqrt{5}}{2} \cong 0.38$$

$$S.S = \{ 2.62, 0.38 \}$$

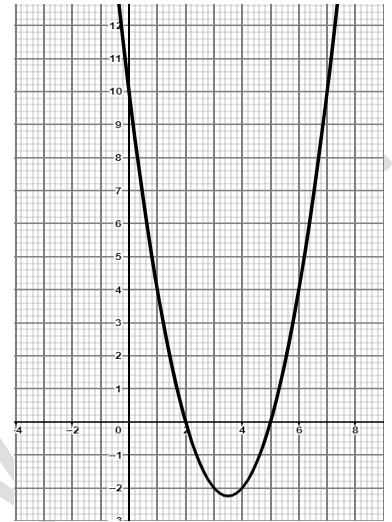
Second: Solving the second degree equation graphicallyExample 7:

Find the solution set of the equation $x^2 - 4 = 0$ in $\mathbb{R}$ graphically.

Solution:

Construct a table contains some values of x and find the corresponding values of y .

x	3	2	1	0	-1	-2	-3
y	5	0	-3	-4	-3	0	5



The solution set is the x coordinate (abscissa) of the intersection points with x -axis: $S.S. = \{2, -2\}$

Note :

we can check the graphical solution by the algebraic solution.

Exercise (1)First:Find the solution set of the following equations:

1. $x^2 = 3x + 18$

2. $x^2 + x - 12 = 0$

3. $x^2 - 25 = 0$

4. $x^2 + 7x - 18 = 0$

5. $x(x - 3) = 5x$

Second:Find the solution set of the following equations by using the general formula:

1. $x^2 + 7x + 2 = 0$

2. $3x^2 - 6x + 1 = 0$

3. $x^2 + 8x + 9 = 0$

Third:Find the solution set of the following equations graphically:

$x^2 - 9 = 0$

2. $x^2 + x - 6 = 0$

Fourth:If $x = -5$ is one of the roots of the equation $x^2 + 2x - m = 0$, then: $m = \dots$

Lesson 2

The complex numbers (C)

Introduction:

Find the solution set of the equation $x^2 + 1 = 0$

Solution:

The equation: $x^2 + 1 = 0$ $x^2 = -1$ (Has no solutions in R)

So it is necessary to study a new set of numbers includes the solution set of this equation.

If the imaginary number $\sqrt{-1} = i$, then:

$$i^2 = -1$$

$$i^3 = -i$$

$$i^4 = 1$$

Notice that:

$$x^2 = -1$$

$$x^2 = i^2$$

$$x = i, -i$$

$$x^2 = -9$$

$$x^2 = 9 i^2$$

$$x = 3 i, -3i$$

➤ These equations have solutions but in the set of the complex numbers

Example (1):

Find in the simplest form:

a) i^6

b) i^{12}

c) i^{17}

d) i^{4n+1}

e) i^{-72}

Solution:

a) $i^6 = i^2 = -1$

b) $i^{12} = 1$

c) $i^{17} = i$

d) $i^{4n+1} = i^{4n} \times i = i$

e) $i^{-72} = 1$

➤ The complex number:

Is the number that can be written in the form

$$z = x + yi \text{ where } x, y \in R, i^2 = -1 \text{ where}$$

" x " is the real part and " yi " is the imaginary part.

➤ Equality of two complex numbers:

If $z_1 = x_1 + y_1i$, $z_2 = x_2 + y_2i$, then: $z_1 = z_2$ if $x_1 = x_2$ and $y_1 = y_2$

Example (2)

Find the values of x and y which satisfy the equation

$$(4x + 1) + 4yi = 5 - 12i$$

Solution:

$$4x + 1 = 5 \quad , \quad 4x = 4 \quad x = 1$$

$$4y = -12 \quad , \quad y = -3$$

Example (3)

Find the values of x and y which satisfy the equation

$$(3x + y) + (x - 4y)i = 5 - 7i$$

Solution:

$$3x + y = 5 \quad , \quad x - 4y = -7$$

(by solving the two simultaneous equations algebraically)

$$x = 1 \quad , \quad y = 2$$

Not :

The calculator can be used to solve two equation algebraically

Exercise (2)

1. Find in the simplest form:

$$i^{73} = \dots, \quad i^{43} = \dots, \quad i^{-54} = \dots, \quad i^{8n+19} = \dots$$

$$i^{-100} = \dots, \quad \frac{1}{i^{15}} = \dots, \quad i^{4n-3} = \dots$$

2. Find the values of a and b which satisfy the equation: $(a+3)-(b-1)i = 7-9i$,

$$i^2 = -1$$

3. Find the values of a and b which satisfy the equation: $12 + 3ai = 4b - 27i$, $i^2 = -1$

4. If $x + yi = 2i$, then: $x = \dots$, $y = \dots$, where $i^2 = -1$

Lesson 3

Operations on the complex numbers

If $z_1 = x_1 + y_1i$, $z_2 = x_2 + y_2i$

- Summation of two complex numbers: $z_1 + z_2 = (x_1 + x_2) + (y_1 + y_2)i$
- Conjugate Numbers:

If $z = x + yi$, then its conjugate symbolized by $\bar{z} = x - yi$

where:

$$z + \bar{z} = x^2 + y^2$$

Example (1):

Put in the simplest form $(4 + 3i) + (2 - i)$

Solution: $(4 + 3i) + (2 - i) = 6 + 2i$

Example (2):

Put in the simplest form $(3 + 5i) - (2 - 3i)$

Solution: $(3 + 5i) - (2 - 3i) = 1 + 8i$

Example (3):

Put in the simplest form $(3 + 2i)(4 - 5i)$

Solution: $(3 + 2i)(4 - 5i) = 12 - 15i + 8i - 10i^2$

$$= 12 - 7i + 10$$

$$= 22 - 7i$$

Example (4):

Put in the simplest form: $z = \frac{3+2i}{4+3i}$

Solution: $z = \frac{3+2i}{4+3i} \times \frac{4-3i}{4-3i} = \frac{12-9i+8i+6}{16+9} = \frac{18}{25} - \frac{1}{25}i$

Example(5):

If $x = \frac{26}{5+i}$, $y = \frac{2(2+3i)}{1+i}$, prove that x and y are conjugate numbers, then: find

a) $x^2 + xy + y^2$

b) $x^2y + xy^2$

solution:

$$\begin{aligned} x &= \frac{26}{5+i} \times \frac{5-i}{5-i} \\ &= \frac{26(5-i)}{25+1} \\ &= 5 - i \end{aligned}$$

$$\begin{aligned} y &= \frac{2(2+3i)}{1+i} \\ y &= \frac{2(2+3i)}{1+i} \times \frac{1-i}{1-i} \\ &= \frac{2(2+3i)(1-i)}{1+1} = 5 + i \end{aligned}$$

$\therefore x$ and y are conjugate numbers

$$\begin{aligned} \text{a) } x^2 + xy + y^2 &= (5-i)^2 + (5-i)(5+i) + (5+i)^2 \\ &= 24 - 10i + 26 + 24 + 10i \\ &= 74 \end{aligned}$$

$$\begin{aligned} \text{b) } x^2y + xy^2 &= xy(x+y) \\ &= (5-i)(5+i)(5-i+5+i) \\ &= 260 \end{aligned}$$

Exercise (3)

1. Find the complex number $(26 - 4i) - (9 - 20i)$ in the simplest form: where $i^2 = -1$
2. Find the simplest form of the expression $(-4i)(-6i)$.
3. The simplest form of the expression $(1 - i)^{10}$ is
4. If $x = (3 + 2i)$, $y = \frac{4-2i}{1-i}$, then: find $x + y$ in the complex number form
5. $x + yi = \frac{2}{1+i}$, then x $y =$
6. Find the values of x and y which satisfy the equation: $\frac{(2+i)(2-i)}{2+4i} = x + yi$

Lesson 4Solving the second degree equation in the complex numbersExample (1):

Solve the following equation: $3x^2 + 12 = 0$

Solution:

$$3x^2 + 12 = 0 \qquad 3x^2 = -12$$

$$x^2 = -4$$

$$x = \pm 2i \text{ S.S} = \{2i, -2i\}$$

Example (2):

Solve the following equation: $2x^2 + 3x + 2 = 0$ by using the general formula

Solution:

$$a = 2, b = -3, c = 2$$

$$x = \frac{-b \pm \sqrt{b^2 - 4ac}}{2a} = \frac{3 \pm \sqrt{9 - 16}}{4}$$

$$= \frac{3 \pm \sqrt{-7}}{4} = \frac{3 \pm \sqrt{7}i}{4}$$

$$\therefore \text{S.S} = \left\{ \frac{3 + \sqrt{7}i}{4}, \frac{3 - \sqrt{7}i}{4} \right\}$$

Exercise (4)

1. Using the general formula solve the equation: $x^2 - 4x + 5 = 0$ in the set of the complex numbers.

2. Using the general formula solve the equation: $x^2 + x + 1 = 0$ in the set of the complex numbers.

3. Solve the following equations in the set of the complex numbers:

a) $5x^2 + 180 = 0$

b) $4y^2 + 20 = 0$

c) $4x^2 + 100 = 75$

Lesson 5

Determining the type of two roots of a quadratic equation

The general form of the second degree equation is : $ax^2 + bx + c = 0$ where a, b and $c \in R, a \neq 0$

3. If the (discriminant) $b^2 - 4ac > 0$, then the two roots are real different numbers.
4. If the (discriminant) $b^2 - 4ac = 0$, then the two roots are equal real numbers.
5. If the (discriminant) $b^2 - 4ac < 0$, then the two roots are two non-real complex numbers conjugate to each other.

Example (1): Determine the type of the two roots for each of the following equations:

a) $x^2 - 4x + 2 = 0$, b) $x^2 - 12x + 36 = 0$, c) $4x^2 + 7x + 5 = 0$

Solution: $x^2 - 4x + 2 = 0$ $a = 1$, $b = -4$, $c = 2$

The discriminant = $b^2 - 4ac$ $= 16 - 8 = 8 > 0$

$\therefore$ the two roots are different real numbers

b) $x^2 - 12x + 36 = 0$

Solution: $x^2 - 12x + 36 = 0$

$a = 1$ $b = -12$ $c = 36$

The discriminant = $b^2 - 4ac = 144 - 144$

$\therefore$ the two roots are equal real numbers

c) $4x^2 + 7x + 5 = 0$

Solution:

$4x^2 + 7x + 5 = 0$

$a = 4$ $b = 7$ $c = 5$

The discriminant = $b^2 - 4ac$ $= 49 - 80 = -31 < 0$

$\therefore$ The two roots are different complex numbers

Example (2):

Find the value of k that makes the two roots of the equation $3x^2 - 6x + k = 0$ are equals.

Solution:

$$a = 3 \quad b = -6 \quad c = k$$

$$\text{The discriminant} = b^2 - 4ac$$

$$36 - 12k = 0$$

$$k = 3$$

Example (3)

The roots of the equation $lx^2 - 12x + 9 = 0$ are complex and not real numbers if

a) $l = 1$

b) $l < 4$

c) $l > 4$

d) $l = 4$

$$a = l, \quad b = -12, \quad c = 9$$

$$b^2 - 4ac < 0$$

$$144 - 36l < 0$$

$$144 < 36l$$

$$4 < l$$

Exercise (5)

1. The discriminant of the equation $ax^2 + bx + c = 0$ is
 2. Determine the type of the two roots of the equation $2x^2 + 3x + 6 = 0$.
 3. If the two roots of the equation $3x^2 - 6x + k = 0$ are equal, then $k = \dots$.
 4. The roots of the equation: $x^2 - 2x + m = 0$ are different complex number, if
- a) $m = 1$ b) $m < 1$ c) $m > 1$ d) $m = 4$

Lesson 6

Relation between the roots of the equation and its coefficients

Let the roots of the equation $ax^2 + bx + c = 0$ be l and m

$$\diamond l + m = \frac{-\text{coefficient of } x}{\text{coefficient of } x^2} = \frac{-b}{a} \qquad l m = \frac{\text{the term free of } x}{\text{coefficient of } x^2} = \frac{c}{a}$$

Important results:

1. If one of the two roots is an additive inverse of the other, then $b = 0$
2. If one of the two roots is a multiplicative inverse of the other, then $a = c$

Example (1)

Complete the following: If $2x^2 + 5x + 3 = 0$, then:

the sum of the two roots =, the product of the two roots =

solution: the sum of the two roots = $\frac{-5}{2}$, the product of the two roots = $\frac{3}{2}$

Example (2)

Complete the following: If one of the two roots of the equation $2x^2 + (a - 5)x + 13 = 0$, is the additive inverse of the other, then $a = \dots$

Solution

$$a - 5 = 0 \quad a = 5$$

Example (3)

complete the following: If one of the two roots of the equation $5x^2 + 26x + a - 2 = 0$, is the multiplicative inverse of the other, then $a = \dots$

Solution

$$a - 2 = 5 \quad \therefore a = 7$$

Example (4)

If one of the roots of the equation $x^2 - 3mx + (7m - 3) = 0$ is twice the other root, find the integer real value of m .

Solution: let the roots of the equation be $l, 2l$

$$\therefore \text{the sum of the roots} = 3l = 3m \qquad \therefore m = l \qquad (1)$$

$$\text{The product of the two roots} = 2l^2 = 7m - 3 \qquad (2)$$

$$\text{From (1) and (2)} \quad \therefore 2m^2 = 7m - 3 \qquad 2m^2 - 7m + 3 = 0$$

$$(2m - 1)(m - 3) = 0 \qquad m = \frac{1}{2} \text{ is refused} \qquad m = 3$$

Exercise (6)

1. If one of the two roots of the equation $x^2 + (a - 5)x - 9 = 0$, is the additive inverse of the other, then: $a = \dots\dots$
2. If one of the two roots of the equation $x^2 + 5x + (l - 4) = 0$ is the multiplicative inverse of the other, then: $l = \dots\dots$
3. If $x = 3$ is one of the two roots of the equation $3x^2 - 2x - E = 0$, then: $E = \dots\dots\dots$
4. If $x = 3$ is one of the two roots of the equation $x^2 + ax - 12 = 0$, then: $a = \dots\dots\dots$, the other root =.....
5. Find the value of m which makes one of the roots of the equation $2x^2 - mx + 3 = 0$ increased from the multiplicative inverse of the other root by 1
6. Find the value of k which makes one of the roots of the equation $x^2 - 6x + k = 0$ equals the square of the other root.
7. If the two roots of the equation $2x^2 + 5x + 4k = 0$ are equal, then find the value of k and the two roots.
8. If the ratio between the two roots of the equation $x^2 - bx + 48 = 0$ equals $3:4$, then find the values of b .

Lesson 7

The sign of the function

❖ First: the constant function:

It's general form is $f(x) = C$ where $C \in R - \{0\}$

Example 1: The sign of the function f where $f(x) = 7$ (is positive)

: The sign of the function f where $f(x) = -3$ (is negative)

❖ Second: The linear function (first degree function)

Its general form is $f(x) = b x + c$ where $b, c \in R, b \neq 0$

- The sign of the function is the same as the sign of b when $x > \frac{-c}{b}$
- The sign of the function is different from the sign of b when $x < \frac{-c}{b}$
- $F(x) = 0$ when $x = \frac{-c}{b}$

Example 2

Determine the sign of the function: $f(x) = 3 x + 6$

Solution:

$$3 x + 6 = 0 \qquad 3 x = -6 \qquad \therefore x = -2$$

The sign of the function is positive when $x > -2$

The sign of the function is negative when $x < -2$

$$f(x) = 0 \quad \text{when} \quad x = -2$$

Example 3

Determine the sign of the function: $f(x) = 5 - x$

Solution:

$$5 - x = 0 \quad \therefore x = 5$$

The sign of the function is negative when $x > 5$

The sign of the function is positive when $x < 5$

$$f(x) = 0 \text{ when } x = 5$$

❖ Third: The quadratic function (second degree function)

Its general form is $f(x) = ax^2 + bx + c$ where $a, b, c \in R, a \neq 0$

1. If the two roots are real different numbers: (L, m)

- The sign of the function is the same as the sign of a when $x \in R - [l, m]$
- The sign of the function is different from the sign of a when $x \in]l, m[$
- $f(x) = 0$ when $x \in \{l, m\}$

Example 4

Determine the sign of the function: $f(x) = x^2 - 3x - 4$, show that on the number line

Solution:

$$x^2 - 3x - 4 = 0 \quad (x - 4)(x + 1) = 0 \quad \therefore x = 4 \text{ or } x = -1$$

The sign of the function is positive when $x \in \mathbb{R} - [-1, 4]$

The sign of the function is negative when $x \in] - 1, 4[$

$$f(x) = 0 \text{ when } x \in \{-1, 4\}$$

The number line:



Example 5:

Determine the sign of the function: $f(x) = 6 - x - x^2$, on $\mathbb{R}$

Solution:

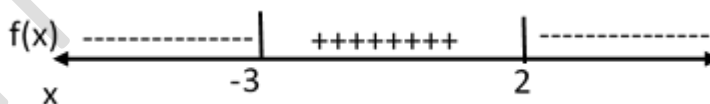
$$6 - x - x^2 = 0 \quad x^2 + x - 6 = 0 \quad (x + 3)(x - 2) = 0 \quad \therefore x = -3 \text{ or } x = 2$$

The sign of the function is negative when $x \in \mathbb{R} - [-3, 2]$

The sign of the function is positive when $x \in] - 3, 2[$

$$F(x) = 0 \text{ when } x \in \{-3, 2\}$$

The number line:



(2) If the two roots are equal real numbers: (L)

- The sign of the function is the same as the sign of a when $x \neq l$
- $f(x) = 0$ when $x = l$

Example (6)

Determine the sign of the function: $f(x) = x^2 - 4x + 4$

Solution:

$$x^2 - 4x + 4 = 0$$

$$(x - 2)(x - 2) = 0 \quad \therefore x = 2$$

The sign of the function is positive when $x \neq 2$

$$f(x) = 0 \text{ when } x = 2$$

(3) If the two roots are not real numbers:

Example 7:

Determine the sign of the function: $f(x) = x^2 - 3x + 5$

Solution:

$$x^2 - 3x + 5 = 0$$

$$b^2 - 4ac = 9 - 20 = -11 < 0$$

The two roots are not real numbers

The sign of the function is positive.

The sign is as same
as the sign of a

Exercise: (7)

1. The function $f: f(x) = 4x - 4$ is negative in the interval
2. The function $y = 3 - x$ is positive in the interval
3. If: $f(x) = 5 - x$, then: $f(x) > 0$ for each x
4. If: $f(x) = -6x$, then: $f(x)$ is positive when $x \in$
5. The sign of the function $f(x) = -7$ is.....
6. If $f(x) = 3 - 2x$, then: $f(x)$ is positive when
7. The sign of the function $f(x) = x^2 + 9$ is positive in the interval.....
8. Determine the sign of the function $f: f(x) = 2x - x^2 + 3$, shown in the number line.
9. Determine the sign of the function $f: f(x) = 1 - 2x + x^2$
10. Determine the sign of the function $f: f(x) = x^2 - 4x + 7$
11. Determine the sign of the function $f: f(x) = 2x^2 + 5x - 3$ where $x \in R$ show that in the number line.
12. Determine the sign of the function $f: f(x) = 12 - 5x - 2x^2$, shown in the number line.
13. Determine the sign of the function $f: f(x) = 3 - x^2 + 2x$, shown in the number line.
14. Determine the sign of the function $f: f(x) = 8 - 2x - x^2$, shown in the number line.
15. Determine the sign of the function $f: f(x) = x^2 - 8x + 16$.

Lesson 8Quadratic inequalities in one variable

You have studied before the first degree inequality in one variable, and you have known that solving the inequality means, finding all values of the unknown which satisfy this inequality, and is written in the form of an interval. Now we are going to find the solution of the second degree inequality in one variable?

Example (1):

find the solution set of the inequality in $\mathbb{R}$ $x^2 + 2x - 8 > 0$

Solution:

Determine the sign of the function $x^2 + 2x - 8 = 0$

$$(x + 4)(x - 2) = 0 \quad \square \quad x = -4 \text{ or } x = 2$$

The sign of the function is positive when $x \in \mathbb{R} - [-4, 2]$

The sign of the function is negative when $x \in] - 4, 2[$

$$f(x) = 0 \text{ when } x \in \{-4, 2\} \quad \text{S.S} = \mathbb{R} - [-4, 2]$$

Example (2):

Find the solution set of the inequality in $\mathbb{R}$ $x^2 - 4x + 4 \geq 0$

Solution:

Determine the sign of the function $x^2 - 4x + 4 = 0$

$$(x - 2)(x - 2) = 0$$

$$\square \quad x = 2$$

The sign of the function is positive when $x \neq 2$

$$f(x) = 0 \text{ when } x = 2$$

$$\square \quad \text{S.S} = \mathbb{R}$$

Example (3):

Find the solution set of the inequality in $\mathbb{R}$ $6x - x^2 - 5 < 0$

Solution:

We investigate the sign of the function: $6x - x^2 - 5 = 0$ multiply by (-1)

$$x^2 - 6x + 5 = 0$$

$$(x - 1)(x - 5) = 0$$

$$\square x = 1 \text{ or } x = 5$$

The sign of the function is negative when $x \in \mathbb{R} - [1, 5]$

The sign of the function is positive when $x \in]1, 5[$

$$F(x) = 0 \text{ when } x \in \{1, 5\}$$

$$\text{S.S} = \mathbb{R} - [1, 5]$$

Example (4):

Find the solution set of the inequality in $\mathbb{R}$ $7x + 15 \leq 2x^2$

Solution:

$$7x + 15 - 2x^2 \leq 0$$

Determining the sign of the function; $7x + 15 - 2x^2 = 0$ multiply by (-1)

$$2x^2 - 7x - 15 = 0 \quad (2x + 3)(x - 5) = 0 \quad \square x = -\frac{3}{2}, \quad x = 5$$

The sign of the function is positive when $x \in \mathbb{R} - [-\frac{3}{2}, 5]$

The sign of the function is negative when $x \in]-\frac{3}{2}, 5[$

$$f(x) = 0 \text{ when } x \in \{-\frac{3}{2}, 5\}$$

$$\text{S.S} = \mathbb{R} -]-\frac{3}{2}, 5[$$

Exercise 8**Find the solution set for each of the following inequalities;**

1. $x^2 - 9 \leq 0$

2. $x^2 - 1 < 0$

3. $2x - x^2 \geq 0$

4. $(x - 1)(x - 3) > 0$

5. $x^2 \geq 4x - 4$

6. $x^2 - 4x + 7 < 0$

Test

Answer the following question

- . If : $f(x) = 4 - 2x$, then the sign of the function f is negative at the interval
- a) $]2, \infty$ b) $]2, 4[$ c) $[2, 4]$ d) $[4, \infty[$

The solution set of the equation: $x^2 + 25 = 0$ in $\mathbb{C}$ is

- a) $[5, -5]$ b) $\{-3i\}$ c) $[5i, -5i]$ d) $\emptyset$

$$1 + i + i^2 + i^3 + i^4 + \dots + i^{16} = \dots \dots \dots$$

- a) i b) 1 c) 16 d) 4

- . If the two roots of the equation $4x^2 - 12x + c = 0$ are equal real numbers, then : $c = \dots$

- a) 3 b) 144 c) 16 d) 9

- . If one of the two roots of the equation $x^2 - (b - 3)x + 5 = 0$ is additive inverse of the other

- a) -5 b) 3 c) 5 d) -3

- . If the two roots of the equation $x^2 + 4x + c = 0$ are real numbers, then $c \in \dots$

- a) $[4, \infty[$ b) $]4, \infty[$ c) $] -\infty, 4]$ d) $] -\infty, 4[$

7. If $(1 + i)(1 - i) = x + iy$, then $x + y = \dots$

- a) 4 b) 3 c) -3 d) 2

8. The solution set of the inequality $x(x - 2) \geq 0$ in $\mathbb{R}$ is

- a) $\{0, 2\}$ b) $[0, 2]$ c) $\mathbb{R} - [0, 2]$ d) $\mathbb{R} -]0, 2[$

- . If one of the two roots of the equation $ax^2 + 2x + 5 = 0$ is multiplicative inverse of the other, then $a = \dots$

- a) -5 b) -2 c) 2 d) 5

Unit Two

Trigonometry

- ❖ *Recognize* the directed Angle.
- ❖ *Recognize* the slandered position of the directed angle.
- ❖ *Recognize* the negative and the positive measure of the directed angle.
- ❖ *Recognize* the radian measure of an angle.
- ❖ *Recognize* the relation between degree measure and radian measure of the directed angle.
- ❖ *Recognize* the trigonometric functions and their reciprocals.
- ❖ Determine the sign of the trigonometric functions in the four quadrants.
- ❖ Deduce the trigonometric functions of some special angles.
- ❖ Know the trigonometric functions.
- ❖ Know how to Solve trigonometric equations.
- ❖ Know the area of the triangle using two sides and included angle.
- ❖ Know the area of the regular polygon using trigonometric functions.

Evaluation

Applied Technology School

Lesson 1

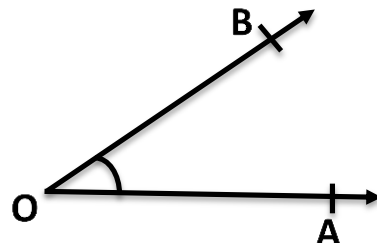
The Directed Angle

Remember that: The angle is union of two rays with the same starting point.

The starting point is called the vertex of the angle.

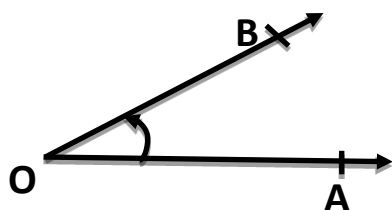
The two rays are called the two sides of the angle.

And it reads as $\angle AOB$ or $\angle BOA$.

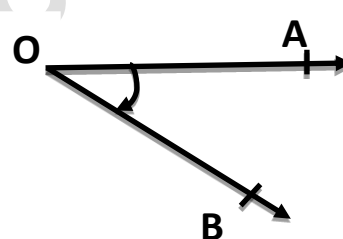


The Directed Angle:

It is an ordered pair of two rays which are called the two sides of the angle with a common starting point which is called the vertex of the angle.



$(\overrightarrow{OA}, \overrightarrow{OB})$ is called directed $\angle AOB$

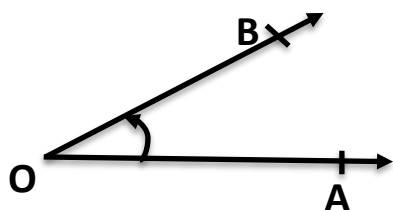


$(\overrightarrow{OB}, \overrightarrow{OA})$ is called directed $\angle BOA$

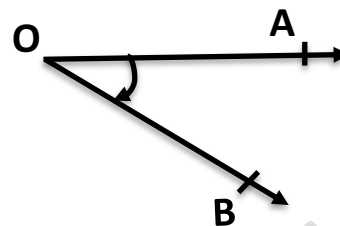
Positive and negative measures of the Directed Angle:

The measure of directed angle is:

- Positive, if the direction of rotation from the initial side to the terminal side is anticlockwise.
- Negative, if the direction of rotation from the initial side to the terminal side is clockwise.



$$m(\angle AOB) = 40^\circ$$

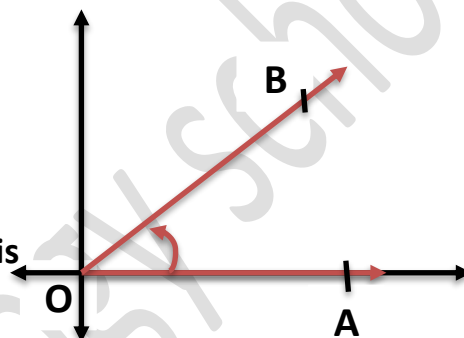


$$m(\angle BOA) = -40^\circ$$

The standard position of the directed angle:

A directed angle is in the standard position if

- 1 - Its vertex is the origin point.
- 2 - Its initial side lies on the positive direction of x-axis



Notes:

- (1) measure of directed $(\angle AOB) \neq$ measure of directed $(\angle BOA)$.
- (2) The directed angle in the standard position has two measures one of them is positive and the other is negative such that the sum of the absolute values two measures $= 360^\circ$.
- (3) The orthogonal coordinate axes divide the plane into four quadrants

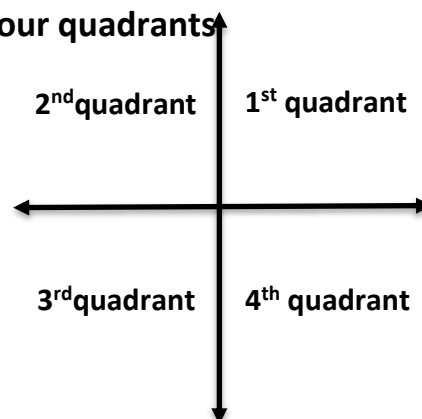
as in the opposite figure:

If $0^\circ < \theta < 90^\circ$ then θ lies in the first quadrant.

If $90^\circ < \theta < 180^\circ$ then θ lies in the second quadrant.

If $180^\circ < \theta < 270^\circ$ then θ lies in the third quadrant.

If $270^\circ < \theta < 360^\circ$ then θ lies in the fourth quadrant.



Equivalent Angles:

All angles whose measure = $\theta \pm n \times 360^\circ$ is equivalent to the angle whose measure = θ

Example 1

The negative measure of the angle whose measure $215^\circ = - (360^\circ - 215^\circ) = -145^\circ$

Example 2

The positive measure of the angle whose measure $(-200^\circ) = 360^\circ - 200^\circ = 160^\circ$

Example 3

The negative measure of the angle whose measure $\frac{\pi}{3} = - (360^\circ - 60^\circ) = -300^\circ$

Example 4

Determine the quadrant in which each of the following angles lies:

(1) 235° (2) -210°

(3) 1270°

Solution

(1) 235°

(third quadrant)

(2) The positive measure of the angle $(-210^\circ) = 360^\circ - 210^\circ = 150^\circ$ (second quadrant).

(3) $1270^\circ = (360 \times 3) - 1270^\circ = 190^\circ$ (third quadrant)

The angle whose measure (1270°) is equivalent to the angle whose measure 190

(third quadrant)

Exercise 1**First question: complete:**

- (1) The measure of the directed angle is positive if the rotation of the terminal side is
- (2) The measure of the directed angle is negative if the rotation of the terminal side is
- (3) The smallest positive measure of the directed angle whose measure 530° is
- (4) The directed angle whose measure 930° lies in the quadrant.
- (5) The smallest positive measure of the directed angle whose measure -690° is

Second question:

Determine the quadrant in which each of the following angles lies:

- (a) 215° (b) -40° (c) -220° (d) 650°

Third question:

Determine a negative measure of each of the following angles:

- (a) 83° (b) 126° (c) 964° (d) 1040°

Lesson 2

Degree Measure and Radian Measure of the Directed Angle

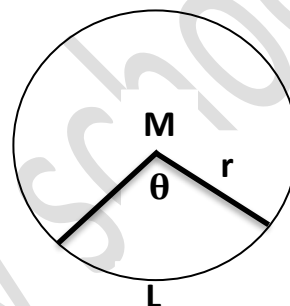
Degree measure: We have known that, the degrees measure is divided into degree minutes and seconds.

Remember that: one degree = 60 minutes, one minute = 60 seconds.

The radian measure of central angle (θ^{rad})

It depends on the arc length (L) and the radius length (r)

$$\theta^{\text{rad}} = \frac{L}{r}$$



To convert from radian measure to degree measure and vice versa

We use the relation

$$\frac{\theta^{\text{rad}}}{\pi} = \frac{\theta^{\circ}}{180^{\circ}}$$

Example 1

Find the radian measure and degree measure of central angle which subtended by an arc of length 14 cm, in a circle whose radius length is 10 cm.

Solution

$$\theta^{\text{rad}} = \frac{L}{r} = \frac{14}{10} = 1.4^{\text{rad}} \quad \theta^{\circ} = \frac{1.4 \times 180}{\pi} = 80^{\circ} 13'$$

Example 2

The arc whose length is 5π cm, in a circle of radius length 15cm, is corresponding to a central angle of measure = $^{\circ}$

Solution

$$\theta^{\text{rad}} = \frac{L}{r} = \frac{5\pi}{15} = \frac{\pi}{3}$$

$$\theta^{\circ} = \frac{180}{3} = 60^{\circ}$$

Note

Substitute with 180° instead of π to find the degree measure

Example 3

The arc length in a circle of diameter length 24 cm and is opposite to a central angle of measure $30^\circ = \dots\dots\dots$ cm

Solution

The radian measure of the central angle whose measure is $30^\circ = \frac{\pi}{6}$

The arc length = $\theta^{\text{rad}} \times r = \frac{\pi}{6} \times 12 = 2\pi$ cm.

Exercise 2

1) A central angle of measure 30° in a circle is subtended by an arc of length 24 cm.

Find the radius length of the circle.

2) The angle whose measure $\frac{5\pi}{6}$ lies in the quadrant.

3) The sum of measures of the interior angles of a triangle in radian measure =

4) The angle whose measure $\frac{7\pi}{3}$ its degree measure = $^\circ$

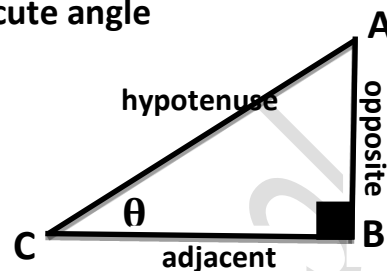
5) Find the radian measure and degree measure of a central angle which subtended by an arc of length 8.7 cm in a circle of radius length 4 cm.

Lesson 3

Trigonometric Functions

We have studied before the basic trigonometric ratios of an acute angle

In the right- angled triangle:



$$\sin \theta = \frac{AB}{AC}$$

$$\cos \theta = \frac{BC}{AC}$$

$$\tan \theta = \frac{AB}{BC}$$

$$\csc \theta = \frac{AC}{AB}$$

$$\sec \theta = \frac{AC}{BC}$$

$$\cot \theta = \frac{BC}{AB}$$

The reciprocals of the basic trigonometric functions

Example 1

ABC is a right- angled triangle at B, AB = 3 cm, BC = 4cm. Find the basic trigonometric functions and its reciprocals of $\angle C$

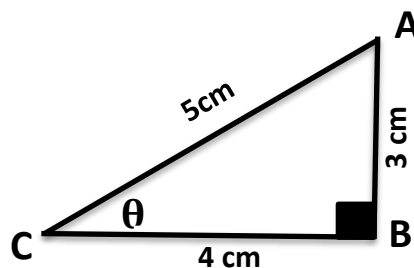
Solution

$$(AC)^2 = (AB)^2 + (BC)^2$$

$$= 9 + 16 = 25$$

$$\therefore AC = 5 \text{ cm}$$

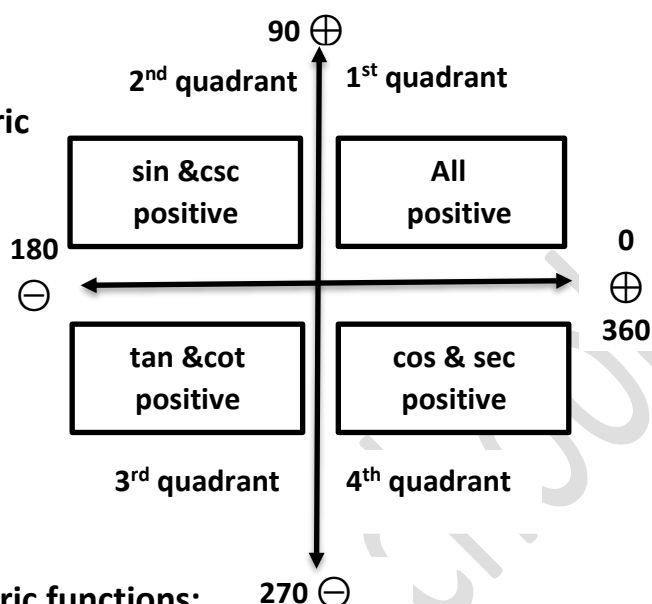
- $\sin C = \frac{3}{5}$ $\csc C = \frac{5}{3}$
- $\cos C = \frac{4}{5}$ $\sec C = \frac{5}{4}$
- $\tan C = \frac{3}{4}$ $\cot C = \frac{4}{3}$



Signs of trigonometric functions:

We can summarize the signs of trigonometric

functions as in the opposite figure

**Example 2**

Write the signs of the following trigonometric functions:

- ❖ $\tan 100^\circ$ (the angle of measure 100° lies in 2nd quadrant) (negative)
- ❖ $\cos 30^\circ$ (the angle of measure 30° lies in 1st quadrant) (positive)
- ❖ $\sin 289^\circ$ (the angle of measure 289° lies in 4th quadrant) (negative)
- ❖ $\tan 869^\circ$ (the angle of measure 869° lies in 2nd quadrant) (negative)
- ❖ $\cos \frac{5\pi}{4}$ (the angle of measure 225° lies in 3rd quadrant) (negative)

Exercise 3

(1) ABC is a right- angled triangle at B in which: $AB = 6$ cm, $AC = 10$ cm. Find the basic trigonometric functions and its reciprocals of $\angle C$

(2) Determine the signs of the following trigonometric functions:

(a) $\cos 210^\circ$ (b) $\sin 740^\circ$ (c) $\tan (-300)^\circ$ (d) $\sin 1230^\circ$

(3) Determine the signs of the following trigonometric functions:

(a) $\sin 240^\circ$ (b) $\tan 365^\circ$ (c) $\csc 410^\circ$

(d) $\cot \frac{9\pi}{4}$ (e) $\sec \left(-\frac{9\pi}{4}\right)$

Lesson 3**The trigonometric functions of some special angles**

The special angles are

$0^\circ, 30^\circ, 45^\circ, 60^\circ, 90^\circ, 180^\circ, 270^\circ$ and 360°

Notes:

- (1) The special angles $0^\circ, 90^\circ, 180^\circ, 270^\circ$ and 360° are called quadrantal angles.
- (2) We can use the calculator to find the values the trigonometric functions of any angle
- (3) The values of the trigonometric functions of the two angles 0° and 360° are as the same.

Example 1

Find the value of: $\sin 30^\circ \cos 60^\circ + \sin 90^\circ - \tan 45^\circ$

Solution

The expression = $0.5 \times 0.5 + 1 - 1 = 0.25$

Example 2

Prove that: $\tan 45^\circ \csc 30^\circ + \cos 60^\circ \csc 30^\circ - \sin 90^\circ \cos 180^\circ = 4$

Solution

L.H.S. = $1 \times 2 + 0.5 \times 2 - 1 \times (-1) = 4 = \text{R.H.S.}$

Exercise 4

(1) Find the values of the following:

(a) $3 \cos 60^\circ - 2 \sin 30^\circ - \cos 270^\circ$

(b) $\sin 30^\circ + \cos^2 45^\circ + \sec^2 45^\circ + \sin^2 90^\circ$

(2) Prove that: $16 \sin^2 60^\circ \cos^2 30^\circ + \tan 45^\circ \cot 45^\circ = 10$

(3) Prove that: $\tan 45^\circ \csc 30^\circ + \cos 60^\circ \csc^2 45^\circ + \sin 90^\circ \cos 0^\circ = 4$

Lesson 5**Related angles**

Two related angles: are two angles, the sum of their measures or the difference between their measures equals a whole number of right angles. θ , $(180 - \theta)$

The trigonometric functions of the two angles whose measures θ and $(180 - \theta)$:

$\sin (180 - \theta) = \sin \theta$	$\csc (180 - \theta) = \csc \theta$
$\cos (180 - \theta) = - \cos \theta$	$\sec (180 - \theta) = - \sec \theta$
$\tan (180 - \theta) = - \tan \theta$	$\cot (180 - \theta) = - \cot \theta$

The trigonometric functions of the two angles whose measures θ and $(180 + \theta)$:

$\sin (180 + \theta) = - \sin \theta$	$\csc (180 + \theta) = - \csc \theta$
$\cos (180 + \theta) = - \cos \theta$	$\sec (180 + \theta) = - \sec \theta$
$\tan (180 + \theta) = \tan \theta$	$\cot (180 + \theta) = \cot \theta$

The trigonometric functions of the two angles whose measures θ and $(360 - \theta)$:

$\sin (360 - \theta) = - \sin \theta$	$\csc (360 - \theta) = - \csc \theta$
$\cos (360 - \theta) = \cos \theta$	$\sec (360 - \theta) = \sec \theta$
$\tan (360 - \theta) = - \tan \theta$	$\cot (360 - \theta) = - \cot \theta$

The trigonometric functions of the two angles whose measures θ and $(- \theta)$:

$\sin (- \theta) = - \sin \theta$	$\csc (- \theta) = - \csc \theta$
$\cos (- \theta) = \cos \theta$	$\sec (- \theta) = \sec \theta$
$\tan (- \theta) = - \tan \theta$	$\cot (- \theta) = - \cot \theta$

Example 1:

Without using calculator, find the value of the following:

$$\cos 210^\circ, \cot (-315^\circ), \sec^2 135^\circ$$

solution

$$\diamond \cos 210^\circ = \cos (180^\circ + 30^\circ) = -\cos 30^\circ = -\frac{\sqrt{3}}{2}$$

$$\diamond \cot (-315^\circ) = -\cot (315^\circ) = -\cot (360^\circ - 45^\circ) = \cot 45^\circ = 1$$

$$\diamond \sec^2 135^\circ = [\sec 135^\circ]^2 = [\sec (180^\circ - 45^\circ)]^2 = [-\sec 45^\circ]^2 = (-\sqrt{2})^2 = 2$$

Example 2:

Without using calculator, find the value of the following:

$$\tan (-225^\circ), \sec 660^\circ, \sin 390^\circ$$

solution

$$\diamond \tan (-225^\circ) = -\tan 225^\circ = -\tan (180^\circ + 45^\circ) = -\tan 45^\circ = -1$$

$$\diamond \sec 660^\circ = \sec 300^\circ = \sec (360^\circ - 60^\circ) = \sec 60^\circ = 2$$

$$\diamond \sin 390^\circ = \sin 30^\circ = \frac{1}{2}$$

Example 3:

Without using calculator, Prove that: $\sin 1125^\circ \cos 585^\circ \csc 870^\circ = -1$

solution

$$\sin 1125^\circ = \sin 45^\circ = \frac{\sqrt{2}}{2}$$

$$\cos 585^\circ = \cos 225^\circ = \cos (180^\circ + 45^\circ) = -\cos 45^\circ = -\frac{\sqrt{2}}{2}$$

$$\csc 870^\circ = \csc 150^\circ = \csc (180^\circ - 30^\circ) = \csc 30^\circ = 2$$

$$\text{The expression} = \frac{\sqrt{2}}{2} \times -\frac{\sqrt{2}}{2} \times 2 = -1$$

Exercise 5

(1) Without using calculator, find the value of the following:

$$\cos 660^\circ, \cos 240^\circ, \sin 390^\circ$$

(2) Without using calculator, find the value of the following:

$$\tan 225^\circ, \csc 510^\circ, \cot 420^\circ$$

(3) Without using calculator, find the value of the following:

$$\sin 150^\circ \cos(-300^\circ) + \cos 930^\circ \cot 240^\circ$$

(4) Prove that:

$$\sin 600^\circ \cos(-30^\circ) + \sin 150^\circ \cos(-240^\circ) = -1$$

Lesson 6**Related angles (follow)**

The trigonometric functions of the two complementary angles whose measures θ and $(90 - \theta)$:

$\sin (90 - \theta) = \cos \theta$	$\csc (90 - \theta) = \sec \theta$
$\cos (90 - \theta) = \sin \theta$	$\sec (90 - \theta) = \csc \theta$
$\tan (90 - \theta) = \cot \theta$	$\cot (90 - \theta) = \tan \theta$

The trigonometric functions of the two angles whose measures θ and $(90 + \theta)$:

$\sin (90 + \theta) = \cos \theta$	$\csc (90 + \theta) = \sec \theta$
$\cos (90 + \theta) = - \sin \theta$	$\sec (90 + \theta) = - \csc \theta$
$\tan (90 + \theta) = - \cot \theta$	$\cot (90 + \theta) = - \tan \theta$

The trigonometric functions of the two angles whose measures θ and $(270 - \theta)$:

$\sin (270 - \theta) = - \cos \theta$	$\csc (270 - \theta) = - \sec \theta$
$\cos (270 - \theta) = - \sin \theta$	$\sec (270 - \theta) = - \csc \theta$
$\tan (270 - \theta) = \cot \theta$	$\cot (270 - \theta) = \tan \theta$

The trigonometric functions of the two angles whose measures θ and $(270 + \theta)$:

$\sin (270 + \theta) = - \cos \theta$	$\csc (270 + \theta) = - \sec \theta$
$\cos (270 + \theta) = \sin \theta$	$\sec (270 + \theta) = \csc \theta$
$\tan (270 + \theta) = - \cot \theta$	$\cot (270 + \theta) = - \tan \theta$

Example 1:

Without using calculator, find the value of the following:

$$\sin 420^\circ \cos (-690^\circ) + \sin 210^\circ \cos 240^\circ$$

solution

- $\sin 420^\circ = \sin 60^\circ = \frac{\sqrt{3}}{2}$
- $\cos (-690^\circ) = \cos 30^\circ = \frac{\sqrt{3}}{2}$
- $\sin 210^\circ = \sin (270^\circ - 60^\circ) = -\cos 60^\circ = -\frac{1}{2}$
- $\cos 240^\circ = \cos (270^\circ - 30^\circ) = -\sin 30^\circ = -\frac{1}{2}$

$$\text{The expression} = \left(\frac{\sqrt{3}}{2}\right) \left(\frac{\sqrt{3}}{2}\right) + \left(-\frac{1}{2}\right) \left(-\frac{1}{2}\right) = 1$$

Example 2:

Without using calculator, find the value of the following:

$$\tan 225^\circ, \quad \csc 510^\circ, \quad \cot 420^\circ$$

solution

$$\tan 225^\circ = \tan (270^\circ - 45^\circ) = \cot 45 = 1$$

$$\csc 510^\circ = \csc 150^\circ = \csc (90 + 60) = \csc 60 = 2$$

$$\cot 420^\circ = \cot 60^\circ = \frac{1}{\sqrt{3}}$$

Exercise 6

(1) Without using calculator, find the value of the following:

$$\sin 600^\circ, \cos 330^\circ, \cos 120^\circ, \sin 150^\circ, \tan(-315^\circ)$$

(2) Without using calculator, Prove that:

$$\tan(-225^\circ) + \sec 660^\circ + \sin 390^\circ = \frac{3}{2}$$

(3) Complete the following:

(a) $\cos(90 + \theta) = \dots\dots\dots$

(b) $\sec(270 + \theta) = \dots\dots\dots$

(c) $\cos(270 - \theta) = \dots\dots\dots$

(d) $\sin 35^\circ = \cos \dots\dots^\circ$

(e) $\tan 42^\circ = \cot \dots\dots^\circ$

(f) $\tan(90 - \theta) = \dots\dots\dots$

LESSON 7**Solving trigonometric equations****Example 1:**

If $0^\circ < x < 90^\circ$, find the value of x which satisfy the equation:

$$\tan (2x + 15) = \cot 27$$

solution

$$\therefore \tan (2x + 15) = \cot 27$$

$$\therefore 2x + 15 + 27 = 90$$

$$\therefore 2x = 90 - 42 = 48$$

$$\therefore x = 24^\circ$$

Example 2:

If $0^\circ < x < 90^\circ$, find the value of x which satisfy the equation: $\cos (2x + 29) = \sin (3x - 64)$

solution

$$\therefore \cos (2x + 29) = \sin (3x - 64)$$

$$\therefore 2x + 29 + 3x - 64 = 90$$

$$\therefore 5x - 35 = 90$$

$$\therefore 5x = 125 \quad \therefore x = 25^\circ$$

Example 3:

If $0^\circ < x < 90^\circ$, find the value of x which satisfy the equation:

$$\cot (30 + x) = \tan (45 + 2x)$$

solution

$$\therefore \cot (30 + x) = \tan (45 + 2x)$$

$$\therefore 30 + x + 45 + 2x = 90$$

$$\therefore 3x + 75 = 90$$

$$\therefore 3x = 15$$

$$\therefore x = 5^\circ$$

Example 4:

If $0^\circ < x < 90^\circ$, find the value of x which satisfy the equation: $\sqrt{2} \sin x = 1$

solution

$$\therefore \sqrt{2} \sin x = 1$$

$$\therefore \sin x = \frac{1}{\sqrt{2}}$$

$$(\text{shift } \sin \frac{1}{\sqrt{2}} =)$$

$$\therefore m(\angle x) = 45^\circ$$

Example 5:

If $0^\circ < x < 90^\circ$, find the value of x which satisfy the equation: $\tan x = 3 \tan^2 30^\circ \sin^2 270^\circ$

solution

$$\tan x = 3 \times \frac{1}{3} \times 1$$

$$\therefore \tan x = 1$$

$$(\text{shift } \tan \frac{1}{\sqrt{2}} =)$$

$$\therefore m(\angle x) = 45^\circ$$

Exercise 7

(1) If $\cot 2\theta = \tan \theta$, where θ is an acute angle, then $m(\angle \theta) = \dots\dots\dots^\circ$

(2) If $\cos \theta = \sin 2\theta$, where θ is an acute angle, then $\sin 3\theta = \dots\dots\dots$

(3) If $0^\circ < x < 90^\circ$, find the value of x which satisfy the equation:

$$\tan x = 4 \sin 30^\circ \cos 60^\circ$$

(4) If $0^\circ < x < 90^\circ$, find the value of x which satisfy the equation:

$$2\sin x = \sin 30^\circ \cos 60^\circ + \cos 30^\circ \sin 60^\circ$$

(5) If $0^\circ < x < 90^\circ$, find the value of x which satisfy the equation:

$$2\cos x = \tan^2 60^\circ - 2\tan 45^\circ$$

LESSON 8**(follow) Solving trigonometric equations****Example 1:**

If $0^\circ < x < 90^\circ$, find the value of x which satisfy the equation:

$$2 \cos x \cos 360 = \csc^2 45 - \cot 45$$

solution

$$2 \cos x \times 1 = 2 - 1$$

$$2 \cos x = 1$$

$$\cos x = \frac{1}{2}$$

(shift $\cos \frac{1}{2} =$)

$$\therefore m(\angle x) = 60^\circ$$

Example 2:

If $0^\circ < x < 90^\circ$, find the value of x which satisfy the equation:

$$4 \sin x \tan \frac{\pi}{4} \cos \frac{\pi}{6} = \sin^2 \frac{\pi}{3} \sec^2 \frac{\pi}{3}$$

solution

$$4 \sin x \times 1 \times \frac{\sqrt{3}}{2} = \frac{3}{4} \times 4$$

$$\sin x = \frac{\sqrt{3}}{2}$$

(shift $\sin \frac{\sqrt{3}}{2} =$)

$$\therefore m(\angle x) = 60^\circ$$

Example 3:

If $x \in] 0^\circ, 90^\circ [$, find the solution set of the equation:

$$\cos 660^\circ \cos 240^\circ + \sin x \sin 390^\circ = 0$$

solution

$$\left(\frac{1}{2}\right)\left(-\frac{1}{2}\right) + \sin x \times \frac{1}{2} = 0$$

$$\therefore \sin x = \frac{1}{2} \quad \left(\text{shift } \sin \frac{1}{2} = \right)$$

$$\therefore m(\angle x) = 30^\circ \therefore \text{S.S.} = \{30^\circ\}$$

Exercise 8

(1) If $0^\circ < x < 90^\circ$, find the value of x which satisfy the equation:

$$\sec x = \tan 225^\circ \csc 510^\circ + \cot 420^\circ \cos 270^\circ$$

(2) If $0^\circ < x < 90^\circ$, find the value of x which satisfy the equation:

$$\cos x \tan 30^\circ = \sin^2 45^\circ$$

(3) If $0^\circ < x < 90^\circ$, find the value of x which satisfy the equation:

$$3 \tan x - 4 \sin 30^\circ = 8 \cos^2 60^\circ$$

(4) If $0^\circ < x < 90^\circ$, find the value of x which satisfy the equation:

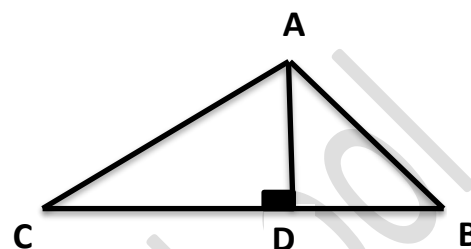
$$3 \sin x = \sin 45^\circ \cos 45^\circ + \sin 30^\circ \cos 60^\circ + \cos^2 30^\circ$$

(5) If x is an acute angle, find the value of x which satisfy the equation:

$$6 \sin x \cos 45^\circ \sin 45^\circ = 1 - \cos^2 60^\circ$$

LESSON 9**The area of the triangle by using two sides and included angle****Remember that:** In any triangle ABC:Area of the triangle = $\frac{1}{2} \times \text{base length} \times \text{height}$

$$= \frac{1}{2} \times BC \times AD$$

**And we will learn in this lesson:****Finding the area of the triangle by using the trigonometric functions**

♦ The area of the triangle =

 $\frac{1}{2}$ the product of length of two adjacent sides $\times$ sine the included angle between them**Example 1:**Find the area of the triangle ABC in which: AB = 11 cm, BC = 10 cm and $m(\angle B) = 47^\circ$ **solution**

$$\text{area of the triangle} = \frac{1}{2} \times AB \times BC \times \sin B$$

$$= \frac{1}{2} \times 11 \times 10 \times \sin 47^\circ = 40.2 \text{ cm}^2$$

Example 2:Find the area of the triangle ABC in which: AB = 6 cm, BC = 8 cm and $m(\angle B) = 90^\circ$ **solution**

$$\text{area of the triangle} = \frac{1}{2} \times AB \times BC \times \sin B$$

$$= \frac{1}{2} \times 6 \times 8 \times \sin 90^\circ = 24 \text{ cm}^2$$

Note: we can find the area of the triangle as

$$\text{area of the triangle} = \frac{1}{2} \times AB \times BC = \frac{1}{2} \times 6 \times 8 = 24 \text{ cm}^2$$

Example 3:

Find the area of the equilateral triangle of side length 12 cm.

solution

$$\begin{aligned} \text{area of the triangle} &= \frac{1}{2} \times 12 \times 12 \times \sin 60^\circ \\ &= 36\sqrt{3} \text{ cm}^2. \end{aligned}$$

Exercise 9

(1) Find the area of the triangle ABC in which: $AC = 12$ cm and the length of the perpendicular

from B to $\overline{AC} = 7$ cm.

(2) Find the area of the triangle ABC in which: $BC = 16$ cm, $BA = 22$ cm and $m(\angle B) = 63^\circ$.

(3) Find the area of the isosceles triangle in which length of each leg is 12 cm and the measure of included angle between them is 64° .

(4) Find the area of the isosceles triangle in which length of one leg is 15 cm and the measure of vertex angle is 60° .

(5) Find the area of the equilateral triangle of side length 20 cm.

Lesson 10**The area of the regular polygon by using trigonometric function****The area of the regular polygon:**

The area of regular polygon of n-sides and side length is (x)

$$= \frac{n}{4} x^2 \cot \frac{\pi}{n}$$

Example 1:

Find the area of the regular octagon of side length 9 cm.

solution

$$\text{area of the regular octagon} = \frac{n}{4} x^2 \cot \frac{\pi}{n}$$

$$= \frac{8}{4} \times 81 \times \cot \frac{180}{8} = 391.1 \text{ cm}^2.$$

Example 2:

Find the area of the equilateral triangle of side length 18 cm.

solution

$$\text{area of the regular triangle} = \frac{n}{4} x^2 \cot \frac{\pi}{n}$$

$$= \frac{3}{4} \times (18)^2 \times \cot \frac{180}{3} = 81\sqrt{3} \text{ cm}^2.$$

Exercise 10

- (1) Find the area of the regular pentagon of side length 15 cm.
- (2) Find the area of the regular hexagon of side length 16 cm.
- (3) Find the area of the regular heptagon of side length 10 cm.
- (4) Find the area of the regular polygon of 12-sides and its side length 6 cm.
- (5) Find the area of the square of side length 4 cm.

Test

First question: Choose the correct answer:

(1) $\tan (90 + \theta) + \cot (180 + \theta) = \dots$

- (a) 0 (b) $2\tan \theta$ (c) $2\cot \theta$ (d) 2

(2) If $\sin \theta > 0$ and $\tan \theta < 0$, then θ lies in the quadrant

- (a) first (b) second (c) third (d) fourth

(3) The arc of length is 5π cm, in a circle of radius length 15cm is opposite to a central angle of measure = $^\circ$.

- (a) 30 (b) 60 (c) 90 (d) 180

(4) If $\sin \theta = \cos 3\theta$, where $\theta \in] 0, 90 [$, then $\sin 4\theta = \dots$

- (a) 0.5 (b) 1 (c) 0 (d) -1

Second question: Complete each of the following

(1) If $2 \sin \theta = \sqrt{2}$, then the smallest positive angle which satisfies that = $^\circ$.

(2) The area of the equilateral triangle of side length 15 cm = cm^2 .

Third question: Complete each of the following

(a) $\cos (90 + \theta) = \dots$

(b) $\sec (270 + \theta) = \dots$

(c) $\cos (270 - \theta) = \dots$

(d) $\cot (180 + \theta) = \dots$

Fourth question:

(1) If $0^\circ < x < 90^\circ$, find the value of x which satisfy the equation:

$$3 \tan x - 4 \sin 30^\circ = 8 \cos 60^\circ$$

(2) Find The area of the regular pentagon of side length 16 cm ?

Evaluation

Applied Technology School